Which coordinates? Match the wording of the question

the question says	*use*	*card*
straight-line motion; a given against t, v or s; v-t, a-t graphs	rectilinear, integrate	Rectilinear
path $y = f(x)$; hill, car, van; thrown ball; x and y move independently	rectangular, projectile	Rectangular and Projectile
curved road, crest or dip, loop, circle; speed along the path; radius ρ	normal-tangential	Normal-Tangential
turning arm, $r(\theta)$, helix, spiral slide, cord pulled through a hole	polar or cylindrical	Polar and Cylindrical
blocks tied by cords over pulleys; cord length is fixed	dependent motion	Dependent Motion
two bodies moving, velocity or acceleration of B as seen from A	relative motion	Relative Motion

Rectilinear Kinematics 12.1-12.3

$v = \dfrac{ds}{dt}, \quad a = \dfrac{dv}{dt}, \quad a\,ds = v\,dv$ last one when $a = f(s)$

$v_{avg} = \dfrac{\Delta s}{\Delta t}, \quad (v_{sp})_{avg} = \dfrac{s_T}{\Delta t}$ s_T = total distance

Constant a_c only:

$v = v_0 + a_c t, \qquad s = s_0 + v_0 t + \frac{1}{2} a_c t^2$

$v^2 = v_0^2 + 2a_c(s - s_0), \qquad s = s_0 + \frac{1}{2}(v_0 + v)t$

Variable acceleration (separate, integrate with limits):

$a = f(t)$: $v = v_0 + \displaystyle\int_0^t a\,dt, \quad s = s_0 + \int_0^t v\,dt$

$a = f(v)$: $t = \displaystyle\int_{v_0}^{v} \frac{dv}{f(v)}, \quad s - s_0 = \int_{v_0}^{v} \frac{v\,dv}{f(v)}$

$a = f(s)$: $\frac{1}{2}(v^2 - v_0^2) = \displaystyle\int_{s_0}^{s} a\,ds$ area under a-s

Terminal speed: set $a = 0$ e.g. $a = g(1 - kv^2)$

$\displaystyle\int \frac{dv}{1 - k^2 v^2} = \frac{1}{2k} \ln \frac{1 + kv}{1 - kv}$ k^2 = the drag k above, so this k in s/m

Graphs and multi-stage motion:

v = slope of s-t, $\quad a$ = slope of v-t

Δv = area under a-t, $\quad \Delta s$ = area under v-t

Stage by stage: the end state of one stage is the start of the next. Two vehicles meet when $s_A = s_B$ at the same t (same origin). Direction reversal at $v = 0$: distance $\neq$ displacement.

Rectangular and Projectile 12.4-12.6

$\mathbf{v} = \dfrac{d\mathbf{r}}{dt}$ (tangent to path), $\quad \mathbf{a} = \dfrac{d\mathbf{v}}{dt}$

$v = \sqrt{v_x^2 + v_y^2 + v_z^2}, \quad a = \sqrt{a_x^2 + a_y^2 + a_z^2}$

Path $y = f(x)$, chain rule:

$v_y = y' v_x, \qquad a_y = y'' v_x^2 + y' a_x$ $y' = dy/dx$

Constant speed v: $v_x = \dfrac{v}{\sqrt{1 + y'^2}}$ hill, van, car problems

Implicit path, e.g. $(y - 40)^2 = 160x$: differentiate both sides in t.

Projectile ($a_x = 0$, $a_y = -g$), joined by the same t:

$x = x_0 + (v_0 \cos\theta_0)\,t$

$y = y_0 + (v_0 \sin\theta_0)\,t - \frac{1}{2} g t^2$

$v_y = v_0 \sin\theta_0 - gt, \quad v_y^2 = (v_0 \sin\theta_0)^2 - 2g(y - y_0)$

Landing on a slope or curve $y = f(x)$: substitute $x(t)$, $y(t)$ into the ground equation. Top of flight: $v_y = 0$.

Normal-Tangential 12.7

$\mathbf{v} = v\,\hat{\mathbf{u}}_t, \qquad \mathbf{a} = a_t \hat{\mathbf{u}}_t + a_n \hat{\mathbf{u}}_n$

$a_t = \dot{v} = v\dfrac{dv}{ds}, \qquad a_n = \dfrac{v^2}{\rho}$ a_n toward centre; ρ in m (ft)

$a = \sqrt{a_t^2 + a_n^2}$

$\rho = \dfrac{[1 + (dy/dx)^2]^{3/2}}{|d^2y/dx^2|}$ crest/trough $y' = 0$: $\rho = 1/|y''|$

$\theta = \tan^{-1}(dy/dx)$ slope; resolve W into t, n

Circle: $v = r\dot{\theta}$, $a_t = r\ddot{\theta}$, $a_n = r\dot{\theta}^2$

$a_t = f(t)$: $v = v_0 + \displaystyle\int a_t\,dt, \quad s = \int v\,dt$ $s = r\theta$ for an arc

Polar and Cylindrical 12.8

$v_r = \dot{r}, \quad v_\theta = r\dot{\theta}, \quad v_z = \dot{z}$ $\dot{\theta}$ in rad/s

$a_r = \ddot{r} - r\dot{\theta}^2, \quad a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}, \quad a_z = \ddot{z}$

$v = \sqrt{v_r^2 + v_\theta^2 + v_z^2}, \quad a = \sqrt{a_r^2 + a_\theta^2 + a_z^2}$

$\dot{r} = \dfrac{dr}{d\theta}\dot{\theta}, \quad \ddot{r} = \dfrac{d^2r}{d\theta^2}\dot{\theta}^2 + \dfrac{dr}{d\theta}\ddot{\theta}$ path $r = f(\theta)$

$r^2 = f(\theta)$: $2r\dot{r} = f'\dot{\theta}$ lemniscate; differentiate implicitly

$\tan\psi = \dfrac{r}{dr/d\theta}$ radial line to tangent

Helix: $\phi = \tan^{-1}\dfrac{\text{lead}}{2\pi r}$, $v_\theta = v\cos\phi$, $v_z = v\sin\phi$

Given $\theta(t)$: find the instant first (e.g. $\theta = 2\pi$ gives t), then all of $r, \dot{r}, \ddot{r}, \dot{\theta}, \ddot{\theta}$ there.

Dependent Motion (pulleys) 12.9

$\Sigma(\text{coeff})\,s_i = l \;\Rightarrow\; \Sigma(\text{coeff})\,v_i = 0, \quad \Sigma(\text{coeff})\,a_i = 0$

Measure each s from a fixed datum, positive the same way. Coefficient = number of cord segments that hang from that block or pulley. More than one cord: one equation per cord, then eliminate.

e.g. $3s_A + s_B = l \;\Rightarrow\; 3v_A = -v_B$ 2022/practice Q4

$\dfrac{d}{dt}\sqrt{x^2 + h^2} = \dfrac{x\dot{x}}{\sqrt{x^2 + h^2}}$ cord to a point at height h

Relative Motion 12.10

$\mathbf{r}_B = \mathbf{r}_A + \mathbf{r}_{B/A}, \quad \mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}, \quad \mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$

Write every vector in $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$ components, then subtract. Answer: magnitude and angle. B on a curve: $\mathbf{a}_B$ has $(a_B)_t$ along $\mathbf{v}_B$ ($-$ if slowing) and $(a_B)_n = v_B^2/\rho$ toward the centre. A straight at constant speed: $\mathbf{a}_A = 0$.

$C^2 = A^2 + B^2 - 2AB\cos c, \quad \dfrac{\sin a}{A} = \dfrac{\sin c}{C}$ triangle method

Units of the easy-to-confuse symbols SI, then FPS

symbol	*meaning*	*SI*	*FPS*
s, x, y, r, l	position, length, cord	m	ft
ρ	radius of curvature	m	ft
v	speed	m/s	ft/s
a, a_t, a_n, g	acceleration	m/s^2	ft/s^2
θ, ϕ, ψ	angle	rad	rad
$\dot{\theta}$, $\ddot{\theta}$	angular rate, accel.	rad/s, rad/s^2	same
lead	helix advance per turn	m	ft
m	mass	kg	slug
W, F, N, T, L	force, weight, lift	N	lb
μ_s, μ_k	friction coefficient	none	none
k (spring)	stiffness	N/m	lb/ft
k in $g(1 - kv^2)$	drag constant	s^2/m^2	s^2/ft^2

slug = lb·s^2/ft. Weight $W = mg$ is a force, so in SI it is N, not kg.

Conversions

1 ft = 12 in = 0.3048 m, 1 mi = 5280 ft, 60 mi/h = 88 ft/s, km/h ÷ 3.6 = m/s, 1 Mm/h = 277.8 m/s, 1 Mg = 1000 kg, 1 lb = 4.448 N, 1 slug = 14.59 kg, 1 hp = 550 ft·lb/s = 745.7 W

Which principle? Decide before you write anything

the question says	*use*	*card*
force or acceleration at one instant	$\Sigma\mathbf{F} = m\mathbf{a}$ with FBD = kinetic diagram	Equations of Motion
speed against position or distance	work and energy	Work and Energy
speed against time, or $F = f(t)$	impulse and momentum	Linear Impulse and Momentum
two bodies push on each other briefly (bullet, coupling)	momentum of the system	Linear Impulse and Momentum
force on a curve (N, T, L)	energy for v^2, then $\Sigma F_n = mv^2/\rho$	Work and Energy, Equations of Motion
orbit, two radii	$r_A v_A = r_B v_B$ plus energy; circular: $v = \sqrt{GM/r}$	Gravitation and Orbits

Traps from past papers: km/h and Mm/h to m/s first. Calculator in radians when θ is in rad. FPS: $m = W/g$. $r = R + h$. Relative answers need magnitude AND angle. Circle the answer.

Equations of Motion 13.1-13.6

$\Sigma\mathbf{F} = m\mathbf{a}$ FBD (forces) = kinetic diagram ($m\mathbf{a}$)

$W = mg, \quad m = W/g$ (slug $= \text{lb}\cdot\text{s}^2/\text{ft}$) mass does not change with g

$F_k = \mu_k N, \quad F_s \le \mu_s N$ friction opposes relative sliding

$F_{sp} = ks$ s = stretch in m (ft), k in N/m (lb/ft)

$\Sigma\mathbf{F}_{ext} = \Sigma m_i \mathbf{a}_i$ system; internal forces cancel

Rectangular:

$\Sigma F_x = ma_x, \quad \Sigma F_y = ma_y$

Crate on accelerating truck: $a_{max} = \mu_s g$ no slip

Pulley systems: one FBD per block, tension same along one cord, plus the cord constraint.

Normal-tangential:

$\Sigma F_t = ma_t, \quad \Sigma F_n = m\,v^2/\rho, \quad \Sigma F_b = 0$

Banked turn/plane: $L\cos\theta = mg,\ L\sin\theta = m\,v^2/r$ $\tan\theta = v^2/(gr)$

Crest: $mg - N = m\,v^2/\rho$ Dip: $N - mg = m\,v^2/\rho$

Inside a loop, at the top: $N + mg = m\,v^2/\rho$ $N = 0$: $v_{min} = \sqrt{g\rho}$

Leaves the surface or cord goes slack: $N = 0$ or $T = 0$.

Cylindrical:

$\Sigma F_r = m(\ddot{r} - r\dot{\theta}^2),\ \Sigma F_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta}),\ \Sigma F_z = m\ddot{z}$

$\Sigma F_\theta = 0 \ \Rightarrow\ r^2\dot{\theta} = \text{const}$ cord pulled through a hole

Spiral slide/coaster: $F_z - mg = m\ddot{z}$, F_r and F_θ from the first two.

Work and Energy 14.1-14.6

$U_{1-2} = \int F\cos\theta\, ds$ + if along motion; J (ft·lb)

$U_W = -W\Delta y$ Δy up positive

$U_s = -\left(\tfrac{1}{2}ks_2^2 - \tfrac{1}{2}ks_1^2\right)$ done BY the spring

Work ON the spring $= +\tfrac{1}{2}k(s_2^2 - s_1^2)$ 2024: s 3 to 1 in gives -4 in·lb

$U_f = -\mu_k N\, d$ N from $\Sigma F_n = 0$ on an incline

$T_1 + \Sigma U_{1-2} = T_2, \quad T = \tfrac{1}{2}mv^2$

Two blocks on a cord: one equation for the whole system, constraint links v_A, v_B and the distances; tension does no net work.

$V_g = Wy, \quad V_e = \tfrac{1}{2}ks^2$ y up from datum

$T_1 + V_1 = T_2 + V_2$ conservative forces only

Coaster limits: $N = 0$ at a crest gives $v^2 = g\rho$; $N = nW$ at a dip gives $v^2 = (n-1)g\rho$; energy then gives h.

$P = \mathbf{F}\cdot\mathbf{v} = Fv, \quad \varepsilon = \dfrac{P_{out}}{P_{in}}$ P in W (ft·lb/s)

Level road, constant v: $F_{friction} = P/v$

$P(t) = F\,v(t),\ v(t)$ from $\Sigma F = ma$ rocket sled, thrust

Pendulum: Poon's method 13.5, 14.6

θ from the horizontal, bob at $\theta = 0$ moving up, cord length l, t along $\mathbf{v}$, n toward the pivot:

ΣF_t: $-mg\cos\theta = ma_t$

ΣF_n: $T + mg\sin\theta = m\,v^2/l$

$v\,dv = a_t\,ds,\ ds = l\,d\theta \ \Rightarrow\ v^2 = v_0^2 - 2gl\sin\theta$ same as energy

At $\theta = 0$: $T_0 = mv_0^2/l$. Slack ($T = 0$): $v^2 = gl\sin\theta$, set equal to the v^2 line: $\sin\theta = v_0^2/(3gl)$. Released from horizontal, at bottom: $v = \sqrt{2gl}$.

Linear Impulse and Momentum 15.1-15.3

$m\mathbf{v}_1 + \Sigma\int_{t_1}^{t_2}\mathbf{F}\,dt = m\mathbf{v}_2$ by components; N·s (lb·s)

$\int F\,dt =$ area under F-t e.g. $\int_0^5 600t^2\,dt$

Starts from rest: moves once $F = \mu_s N$ at t_1 integrate from t_1, use $\mu_k N$

Count the cable strands pulling on the body (2 strands: $2T$).

$\Sigma m_i(v_i)_1 = \Sigma m_i(v_i)_2$ no external impulse along that axis

$v = \dfrac{m_A v_A + m_B v_B}{m_A + m_B}$ bodies move together after

$\Delta T = \Sigma T_1 - \Sigma T_2$ energy lost when bodies join

During a short impact, weight and friction are non-impulsive: leave them out.

Bullet through a block (2024 and later sittings):

1) x: $m_b v_{b1x} + 0 = m_b v_{b2x} + m_B v_B$. Use slope triangles (12/13, 4/5).

2) y, if asked for avg N: $m_b(v_{b1})_y + (N - W_B)\Delta t = m_b(v_{b2})_y$, signs from the figure.

3) Slide to rest: $m_B v_B - \mu_k W_B t = 0 \ \Rightarrow\ t = \frac{v_B}{\mu_k g}$.

$F_{avg} = \dfrac{m\,\Delta v}{\Delta t}$

Gravitation and Orbits 13.7, 14.6

$F = \dfrac{GMm}{r^2}, \quad GM = gR^2$ $R = 6370$ km; $GM = 3.976(10^{14})$ m^3/s^2

$r = R + h$ altitude h is given, not r

Circular orbit ($\Sigma F_n = mv^2/r$)**:**

$v = \sqrt{\dfrac{GM}{r}}, \quad vT = 2\pi r \ \Rightarrow\ r = \left(\dfrac{GMT^2}{4\pi^2}\right)^{1/3}$

Geosynchronous: $T = 23.934$ h $= 86\,162$ s $r \approx 42.1$ Mm

Ellipse, A and B at perigee/apogee ($\mathbf{v} \perp \mathbf{r}$)**:**

$r_A v_A = r_B v_B$ central force: $h = r^2\dot{\theta} = rv_\theta$ const

$\tfrac{1}{2}v_A^2 - \dfrac{GM}{r_A} = \tfrac{1}{2}v_B^2 - \dfrac{GM}{r_B}$ energy, $V_g = -GMm/r$

$v_B^2 = v_A^2 + 2GM\left(\dfrac{1}{r_B} - \dfrac{1}{r_A}\right)$ any two points

$v_A = \sqrt{\dfrac{2GM\,r_B}{r_A(r_A + r_B)}}$ both equations combined

$v_e = \sqrt{\dfrac{2GM}{r}}$ escape speed

$G = 66.73(10^{-12}),\ M_e = 5.976(10^{24})$ kg when G, M given

KE of a satellite: $T = \tfrac{1}{2}mv^2$; the local g does not enter.

Units of the easy-to-confuse symbols SI, then FPS

symbol	*meaning*	*SI*	*FPS*
U, V, T	energies (T = kinetic)	J = N·m	ft·lb
P, ε	power; efficiency (none)	W = J/s	ft·lb/s
$\int F\,dt$, $m\mathbf{v}$	impulse, momentum	N·s	lb·s
F_{avg}	average force	N	lb
R, r, h	radii, altitude	m, not km	ft
GM	gravitational parameter	m^3/s^2	ft^3/s^2
G	gravitational constant	m^3/(kg·s^2)	
M_e	Earth mass	kg	
$r^2\dot{\theta} = rv_\theta$	momentum per mass	m^2/s	ft^2/s
T (orbit)	**period**	s	s

T is kinetic energy (J), tension (N) or period (s): read the context. h is altitude here but $r^2\dot{\theta}$ in the Ellipse card is angular momentum per mass. N·s = kg·m/s; lb·s = slug·ft/s. Convert km, Mm, h to m, s before using GM.